VAIBHAV JAYSING YENARE

Visiting Research Student, TIFR Mumbai • Assistant Professor (ad hoc), B. K. Birla College

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EDUCATION

M.Sc. in Physics, specialization in Astrophysics Jul 2023 – Mar 2025

St. Xavier's College (Empowered Autonomous), Mumbai, India

- CGPA: **8.35 / 10** (First Class)
- Thesis: *Identification of White Dwarf–Main Sequence Binary Systems using UV–Optical Observations*; advisor: Prof. Sourav Chatterjee (TIFR) [report]

B.Sc. in Physics, minors in Mathematics & Statistics Jun 2020 – Apr 2023

St. Xavier's College (Empowered Autonomous), Mumbai, India

- CGPA: **7.23 / 10**

MANUSCRIPTS IN PREPARATION

1. **Yenare, V.**, Chatterjee, S., Ganguly, A., & Nayak, P. K. *White Dwarf–Main Sequence Binaries Using Multi-band Photometry: Discovery and Characterisation.* To be submitted to *The Astrophysical Journal*.
2. Patel, C., Akbari, K., Nadar, R., **Yenare, V.**, Choudhury, M., et al. *Temporal and Spectral Evolution of 4U 1630–47 During the RXTE Era.* To be submitted to *Journal of High Energy Astrophysics*.

RESEARCH EXPERIENCE

Visiting Research Student, Tata Institute of Fundamental Research (TIFR), Mumbai Nov 2024 – Present

Advisor: Prof. Sourav Chatterjee

Collaborators: Dr. A. Ganguly (IUCAA, Pune), Dr. P. K. Nayak (IISER Pune)

Finding white dwarfs hidden on the main sequence — began as M.Sc. thesis; continued after graduation.

- Identified **5,468 white dwarf–main sequence binaries**, **5,099 of them new** relative to six earlier catalogs, by selecting ultraviolet excess (GALEX GR6+7) in stars on the optical main sequence (*Gaia* DR3).
- Flagged **306 extremely low-mass (ELM) white dwarf candidates** ($\lesssim 0.3\ M_\odot$) as targets for spectroscopic follow-up; derived white dwarf masses for 3,213 systems (0.2–1.3 $M_\odot$) from cooling tracks.
- Built a magnitude-limited sample in which almost all systems lie beyond 100 pc and over half beyond 500 pc, reaching a population inaccessible to volume-limited 100 pc surveys.
- Designed colour–magnitude selection cuts from 133,731 quality-filtered sources after standard criteria excluded viable candidates; developed a two-step SED fit in VOSA (Koester and Kurucz atmospheres) that constrains metallicity from the companion before fitting both stars jointly.
- Validated white dwarf temperatures against five independent catalogs (agreement below ~20,000 K).

Research Project, St. Xavier's College, Mumbai Jan 2025 – Present

Advisor: Prof. Manojendu Choudhury

Timing and spectral evolution of the black hole X-ray binary 4U 1630–47

- Analyze the full RXTE archive to trace the joint timing and spectral evolution across every outburst cycle.
- Identified statistically significant Spearman rank correlations between spectral and timing parameters, linked to the appearance of quasi-periodic oscillations (QPOs).

Short research projects (M.Sc., St. Xavier's College)

- **Spin Estimation of the M33 X-8 Black Hole by Spectrum Analysis** — NuSTAR spectral fitting. [report]
- **Molecular Spectral Line Analysis of SiS in IRC +10216** — VLA data reduction in CASA. [report]
- **Photometric Data Analysis for Estimating Galaxy Parameters** — Bayesian SED fitting with Bagpipes. [report]
- **Luminosity Distance Estimation of BH–BH Mergers** — Bayesian inference on GW data with Bilby. [report]
- **Probing AGN Activity in NGC 660** — e-MERLIN/EVN radio imaging and X-ray spectroscopy. [report]

TEACHING EXPERIENCE

Assistant Professor (ad hoc), Department of Physics Aug 2025 – Present
B. K. Birla College (Empowered Autonomous), Kalyan, India

- Teach six courses to 60 undergraduates: Quantum Mechanics, Electrodynamics, Thermodynamics, Special Theory of Relativity, Python Programming and Numerical Methods.
- Teach an introductory Astronomy & Astrophysics certificate course open to students of all years.
- Supervised 6 undergraduate short research projects; contribute to assessment design and academic mentoring.

PRESENTATIONS

2026 *Tracing White Dwarf–Main Sequence Binaries Using Multi-band UV–Optical Photometry.* 44th Annual Meeting of the Astronomical Society of India, IIT Guwahati, May 15–19 Poster

HONORS & AWARDS

2026 Selected for a poster presentation at the 44th Annual Meeting of the Astronomical Society of India (ASI) for the project titled *Tracing White Dwarf–Main Sequence Binaries Using Multi-band UV–Optical Photometry*
2024 First Prize, PROBE 2024 poster competition, St. Xavier's College (department-wide, UG and PG)
2022 First Prize, JIGYASA 2022 poster competition, St. Xavier's College (department-wide, UG and PG)

SCHOOLS & WORKSHOPS

2024 Gravitational Wave Open Data Workshop (GW ODW 2024), completed Apr 30, 2024 [certificate]

TECHNICAL SKILLS

Programming	Python, Bash, SQL, LaTeX
Python	Astropy, galpy, mwdust, Bagpipes, Bilby, GWpy, EinsteinPy, scikit-learn, OpenCV
Astro tools	VOSA, HEASoft (XSPEC), CIAO, CASA, CARTA, MESA, SAOImage DS9
Surveys	GALEX, *Gaia*, Pan-STARRS, 2MASS, WISE
Facilities	RXTE, NuSTAR, VLA, e-MERLIN, EVN; gravitational-wave open data (GWOSC)

OUTREACH & SERVICE

Organizer, Stargazing Nights, B. K. Birla College Aug 2025 – Present

- Organize public stargazing evenings (30–60 attendees each), opening each with a short astrophysics talk.
- Deliver talks for stargazing organizations on how stars live and die, for audiences without a physics background.

Co-founder & Coordinator, Astrophysics Discussion Club, St. Xavier's College Aug 2023 – Mar 2025

- Founded and led journal-club discussions of current astrophysics papers and methods for about 10 members.

Science Communicator, Horizon Astronomical Society Dec 2023 – Mar 2025

- Conducted public stargazing sessions, operated telescopes and explained astronomy to general audiences.

ENGLISH PROFICIENCY

TOEFL iBT **100 / 120** (band 5.0 / 6, CEFR C1), Sep 2026; medium of instruction for B.Sc. and M.Sc.: English.

REFERENCES

Prof. Sourav Chatterjee
Associate Professor
TIFR, Mumbai
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M.Sc. thesis & research advisor

Prof. Manojendu Choudhury
Assistant Professor
St. Xavier's College, Mumbai
manojendu.choudhury@xaviers.edu
M.Sc. instructor & project advisor

Dr. Prasanta K. Nayak
Assistant Professor
IISER Pune
nayakphy@gmail.com
Collaborator (WD–MS binaries)